

Science — Cbse Annual Exam — 80 Marks

Subject: Science | Class: Class 9

Total Marks: 80 | Time Allowed: 3h | CBSE

General Instructions:

1. Read all questions carefully before answering.
2. All questions are compulsory unless stated otherwise.
3. Marks for each question are indicated on the right.
4. Write neatly and legibly.

Q.NO.	QUESTIONS	MARKS
Section A Attempt all 20 questions. Each carries 1 mark(s).		
1	What are skeletal muscle fibres described as? (A) Spindle-shaped, with a single nucleus, lack striations (B) Cylindrical, branched, with a single nucleus, faint striations (C) Long, cylindrical, unbranched, multinucleate, striated (D) Short, branched, multinucleate, lack striations <i>[Ch 3: Tissues in Action]</i>	1
2	Why are weather forecasts less certain further into the future? (A) Scientists do not collect enough data for long-term forecasts (B) Tiny differences in conditions can grow over time and lead to completely different outcomes (C) Measuring instruments are not accurate enough for long-range forecasts (D) Weather patterns repeat in unpredictable cycles <i>[Ch 1: Exploration: Entering the World of Secondary Science]</i>	1
3	Assertion (A): Calcium carbonate deposits around hot springs may have helped in the formation of the first protective membrane. Reason (R): These deposits protected early organic molecules from harmful radiation and extreme conditions. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 2: Cell: The Building Block of Life]</i>	1



Q.NO.	QUESTIONS	MARKS
4	Assertion (A): Experiments by Louis Pasteur proved that new life always comes from pre-existing life. Reason (R): For long, people thought living things arose spontaneously from non-living matter. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 11: Reproduction: How Life Continues]</i>	1
5	In a sound wave, what is a compression? (A) A region where density is higher than the average density (B) A region where density is lower than the average density (C) A region where density is equal to the average density (D) A region where no particles exist <i>[Ch 10: Sound Waves: Characteristics and Applications]</i>	1
6	Why do vegetatively propagated plants tend to be more vulnerable to diseases than sexually reproduced plants? (A) They grow at a slower rate than sexually reproduced plants (B) They produce fewer flowers and fruits than sexually reproduced plants (C) They cannot absorb nutrients from the soil efficiently (D) They have less genetic diversity, so a single disease can affect all the plants <i>[Ch 11: Reproduction: How Life Continues]</i>	1
7	In the hierarchical classification system, which category comes immediately after Class? (A) Order (B) Family (C) Genus (D) Phylum <i>[Ch 12: Patterns in Life: Diversity and Classification]</i>	1
8	When atoms combine, the total energy of the system becomes lower, making the arrangement more stable. What is the force that holds these atoms together called? (A) Nuclear force (B) Gravitational force (C) Magnetic force (D) Chemical bond <i>[Ch 9: Atomic Foundations of Matter]</i>	1



Q.NO.	QUESTIONS	MARKS
9	Which organelle is continuous with the outer membrane of the nuclear envelope? (A) Golgi apparatus (B) Endoplasmic reticulum (C) Mitochondria (D) Lysosome <i>[Ch 2: Cell: The Building Block of Life]</i>	1
10	What does an object do when a net force acts on it, according to Newton's second law of motion? (A) Accelerates opposite to the direction of the net force (B) Accelerates in the direction of the net force (C) Moves with constant velocity in the direction of the net force (D) Rotates perpendicular to the net force <i>[Ch 6: How Forces Affect Motion]</i>	1
11	Assertion (A): The intermediate points on a straight line position-time graph represent possible values for the position of a vehicle at intermediate times if the vehicle is moving with a constant speed. Reason (R): A position-time graph only represents the positions of an object at the specific instants of time for which data was recorded. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 4: Describing Motion Around Us]</i>	1
12	Assertion (A): Two points separated by 0.05 mm cannot be seen as distinct by the unaided human eye from a distance of 25 cm. Reason (R): The limit of resolution of the human eye is 0.1 mm. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 2: Cell: The Building Block of Life]</i>	1
13	An object is said to be at rest if: (A) Its speed is zero. (B) Its position with respect to the reference point does not change with time. (C) Its average velocity is zero. (D) It does not travel any distance. <i>[Ch 4: Describing Motion Around Us]</i>	1



Q.NO.	QUESTIONS	MARKS
14	What does the magnifying glass symbolise in the context of scientific exploration? (A) Mathematical precision in calculations (B) Careful observation and noticing patterns (C) Direction in choosing appropriate models (D) Experimental testing of hypotheses <i>[Ch 1: Exploration: Entering the World of Secondary Science]</i>	1
15	Assertion (A): Mature red blood cells in humans cannot divide or repair themselves. Reason (R): Mature red blood cells lack a nucleus. (A) Both A and R are true, and R is the correct explanation of A. (B) Both A and R are true, but R is NOT the correct explanation of A. (C) A is true but R is false. (D) A is false but R is true. <i>[Ch 2: Cell: The Building Block of Life]</i>	1
16	A girl carries a box while walking on a horizontal surface. No work is done by the upward force she applies on the box because: (A) The force applied is zero (B) The displacement of the box is zero (C) The mass of the box is zero (D) The force is perpendicular to the displacement of the box <i>[Ch 7: Work, Energy, and Simple Machines]</i>	1
17	Which scientist reported that all animals are made up of cells? (A) Matthias Schleiden (B) Theodor Schwann (C) Rudolf Virchow (D) Gottlieb Haberlandt <i>[Ch 2: Cell: The Building Block of Life]</i>	1
18	The principle of purification by crystallization is based on: (A) The difference in densities of substances (B) The difference in boiling points of substances (C) The difference in particle sizes of substances (D) The differences in the solubility of a substance at different temperatures <i>[Ch 5: Exploring Mixtures and their Separation]</i>	1



Q.NO.	QUESTIONS	MARKS
19	In Bohr's model of the atom, what are the fixed circular paths in which electrons revolve called? (A) Parmanus (B) Stationary states or shells (C) Nucleus (D) Dyads <i>[Ch 8: Journey Inside the Atom]</i>	1
20	What happens when excess atmospheric CO₂ is absorbed by the oceans? (A) Sea water becomes more acidic, threatening plankton and coral reefs (B) Sea water becomes more alkaline, promoting coral reef growth (C) Sea water temperature decreases, increasing marine biodiversity (D) Sea water absorbs more oxygen, benefiting fish populations <i>[Ch 13: Earth as a System: Energy, Matter, and Life]</i>	1
Section B Attempt 5 out of 7.		
21	How can one scientifically disprove the claim that food becomes harmful during an eclipse? <i>[Ch 1: Exploration: Entering the World of Secondary Science]</i>	2
	OR Fill in the blanks. Among magnesium and chlorine, magnesium atom can give two electrons to become Mg²⁺. However, chlorine can take only one electron to become _____. Now, _____ ion of magnesium and _____ ions of chlorine combine to give magnesium chloride. <i>[Ch 9: Atomic Foundations of Matter]</i>	
22	In F.C. Steward's 1958 experiment, what process allows mature phloem cells to form an undifferentiated mass of unspecialised cells before redifferentiating? <i>[Ch 3: Tissues in Action]</i>	2
	OR At what minimum difference in boiling points can two miscible liquids be separated by distillation? <i>[Ch 5: Exploring Mixtures and their Separation]</i>	
23	Differentiate between Rough Endoplasmic Reticulum (RER) and Smooth Endoplasmic Reticulum (SER) based on their structure and primary function. <i>[Ch 2: Cell: The Building Block of Life]</i>	2
24	An object moves with a constant velocity of 20 m s⁻¹ for 6 s. Calculate its displacement using the area under the velocity-time graph. <i>[Ch 4: Describing Motion Around Us]</i>	2



Q.NO.	QUESTIONS	MARKS
25	What does the force of friction depend upon? How can a spring balance be used to measure it? <i>[Ch 6: How Forces Affect Motion]</i>	2
Section C Attempt 5 out of 7.		
26	40 g of salt is dissolved in 160 g of water. Calculate the mass by mass percentage of the solution formed. <i>[Ch 5: Exploring Mixtures and their Separation]</i>	3
	OR How does the epidermis protect plants from various environmental factors, and what role does the cuticle play in plants living in dry habitats? <i>[Ch 3: Tissues in Action]</i>	
27	Differentiate between osmosis and diffusion based on their definitions and the requirement of a membrane. <i>[Ch 2: Cell: The Building Block of Life]</i>	3
	OR A stack of coins is pushed by a stretched rubber band and released on a wooden table and then on a polished marble floor. On which surface will the coins travel a larger distance before stopping, and why? <i>[Ch 6: How Forces Affect Motion]</i>	
28	What is a scientific model? Explain with two examples how models simplify real systems by deliberately ignoring certain details. <i>[Ch 1: Exploration: Entering the World of Secondary Science]</i>	3
29	If a marble moving along the inner boundary of a circular ring is released by lifting the ring, in what direction does it continue to move? What does this indicate about the direction of its velocity at any instant during circular motion? <i>[Ch 4: Describing Motion Around Us]</i>	3
30	Does the potential energy of an object near the surface of the Earth change if it moves with constant velocity in the horizontal direction? What if the object is gradually raised in the vertical direction? <i>[Ch 7: Work, Energy, and Simple Machines]</i>	3
Section D Attempt 4 out of 6.		
31	A student said, "The Earth moves around the Sun". In this context, discuss whether an object kept on the Earth can be considered to be at rest. <i>[Ch 4: Describing Motion Around Us]</i>	5



Q.NO.	QUESTIONS	MARKS
	OR Differentiate between laws, theories, and principles in science. Explain each term with an example and discuss why a theory in science is not the same as a guess. <i>[Ch 1: Exploration: Entering the World of Secondary Science]</i>	
32	Describe how a rocket lifts off the ground based on Newton's third law of motion. Also, explain how a spacecraft moving in space where gravitational force is negligible can slow down. <i>[Ch 6: How Forces Affect Motion]</i>	5
	OR Define totipotency. Describe how F.C. Steward demonstrated this property using carrot phloem cells, and explain the processes of dedifferentiation and redifferentiation that occur during the regeneration of a complete plant. <i>[Ch 3: Tissues in Action]</i>	
33	Explain the process of osmosis and how it differs from diffusion. Describe the behaviour of cells when placed in isotonic, hypotonic, and hypertonic solutions, using the example of potato pieces in plain water and salt solution. <i>[Ch 2: Cell: The Building Block of Life]</i>	5
34	Describe the process of paper chromatography for separating the components of a mixture. Explain the principle behind this separation and mention how it can be applied to separate pigments from plant materials. <i>[Ch 5: Exploring Mixtures and their Separation]</i>	5
Section E Attempt 3 out of 4.		
35	Case Study: Cell Membrane and Osmosis A student set up an experiment with two potato pieces of roughly equal size and equal initial weight. One piece was placed in Beaker A containing plain water, and the other in Beaker B containing a 20 per cent salt solution. After an hour, the piece in Beaker A swelled and its weight increased, while the piece in Beaker B shrank and its weight decreased. The student learned that the cell membrane is extremely thin, about 7 to 10 nm thick, and is described by the fluid-mosaic model. In this model, the membrane has a lipid bilayer with proteins embedded in them acting as gatekeepers, and the molecules can move sideways, flip, and rotate within the membrane. (i) Why did the potato piece in Beaker A swell when placed in plain water? (ii) What is the role of proteins embedded in the lipid bilayer of the cell membrane? (iii) Explain the process that caused the potato piece in Beaker B to shrink and lose weight when placed in a 20 per cent salt solution. (iv) How does the fluid-mosaic model explain the 'fluid' nature of the cell membrane? <i>[Ch 2: Cell: The Building Block of Life]</i>	5



Q.NO.	QUESTIONS	MARKS
	OR Case Study: Kinematic Equations for Vehicles on a Straight Road A motorbike is moving on a straight road with an initial velocity of 28 m/s. The rider applies brakes and the motorbike decelerates uniformly, coming to rest after travelling a distance of 98 m. On a parallel track, a car starts from rest and accelerates uniformly, reaching a final velocity of 24 m/s in 6 seconds. A physics student watching both vehicles decides to compare their accelerations and the distances they cover using the kinematic equations for motion in a straight line with constant acceleration. (i) What is the average acceleration of the car during the 6 seconds? (ii) Using the equation $v^2 = u^2 + 2as$, calculate the acceleration of the motorbike. (iii) Calculate the distance travelled by the car in 6 seconds and the time taken by the motorbike to come to a stop. (iv) Compare the magnitudes of acceleration of the motorbike and the car. Which one is greater? <i>[Ch 4: Describing Motion Around Us]</i>	
36	Case Study: Building Scientific Models A student wants to model how long it takes for a ball thrown upward to return to the ground. She considers several factors: the mass of the ball, its initial speed and direction, air resistance, the colour of the ball, and the spin of the ball. She recalls that physicist Meghnad Saha simplified the study of stars by ignoring complex processes and focusing only on temperature, pressure, and how atoms formed ions, which helped explain the connection between star colour and temperature. She decides to apply a similar simplification approach. She also notes that science uses precise symbols like m for mass, v for velocity, and F for force, and that equations are compact statements about relationships between quantities. (i) What three factors did Meghnad Saha focus on to explain the colour of stars? (ii) Why should the student ignore the colour of the ball when modeling the time it takes to return to the ground? (iii) The student ignores air resistance and spin in her simple model. Later, she wants a more accurate prediction. What should she do, based on the approach described for building complex models? (iv) The student writes an equation relating distance, time, and velocity to describe the ball's motion. Explain why this equation is more than just a calculation tool. <i>[Ch 1: Exploration: Entering the World of Secondary Science]</i>	5



Q.NO.	QUESTIONS	MARKS
37	Case Study: Connective Tissues and Blood Components Rohan, a 15-year-old student, was playing football when he fell and scraped his knee. He noticed red blood oozing from the cut, which eventually formed a clot. Upon examining his elbow, he felt a hard, rigid structure. The doctor explained that both blood and bones are connective tissues, but they differ in their matrix — blood has a watery, soft, jelly-like matrix while bone has a hard, rigid matrix containing calcium and phosphorus compounds. The doctor also noted that at the injured site, white blood cells collected, causing inflammation and redness. (i) What gives blood its red colour? (ii) Why do blood and bone differ in consistency despite both being connective tissues? (iii) If platelets were absent in Rohan's blood, what problem would occur at the site of his cut? What role do white blood cells play at the injury site? (iv) Both blood and bone are classified as connective tissues. How does their differing matrix composition relate to their specific functions in the body? <i>[Ch 3: Tissues in Action]</i>	5

